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***B.Tech. Degree III Semester Supplementary Examination
November 2020***

**ME 15-1303 MECHANICS OF SOLIDS
(2015 Scheme)**

Time: 3 Hours

Maximum Marks: 60

**PART A
(Answer ALL questions)**

(10 × 2 = 20)

- I. (a) Draw the stress strain diagram of a typical ductile material and mark the salient points.
 (b) Explain the significance of Mohr's circle and its uses.
 (c) Differentiate between plane stress and plane strain.
 (d) List the assumptions used in torsion equation.
 (e) What do you mean by point of contra flexure? Is the point of contra flexure and point of inflection different?
 (f) What do you understand by neutral axis and moment of resistance? How do you locate neutral axis?
 (g) State and explain any one theory of failure applicable for cast iron.
 (h) Sketch and explain the shear stress distribution in a symmetrical I section.
 (i) What is meant by equivalent length of a column? What are its values under typical loading conditions?
 (j) Discuss the effect of the slenderness ratio of a column over buckling load.

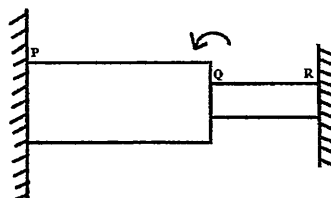
PART B

(4 × 10 = 40)

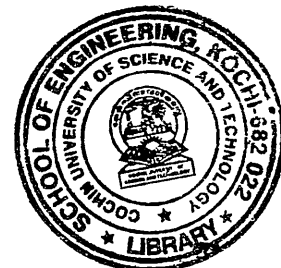
- II. A compound tube consists of steel tube 170 mm external diameter and 10 mm thickness and an outer brass tube 190 mm external diameter and 10 mm thickness. The two tubes are of equal length. The compound tube carries an axial load of 1 MN. Find the stresses and the load carried by each tube and the amount by which it shortens. Length of each tube is 200 mm
 $E_{\text{Steel}} = 200 \text{ GN/m}^2$ and $E_{\text{Brass}} = 100 \text{ GN/m}^2$.

OR

- III. Calculate the thickness of metal required for a C.I pipe of 800 mm diameter, to transport water under a static head of 100 m, if the permissible tensile stress is 20 MPa.
- IV. Two solid shafts PQ and QR of aluminium and steel respectively are rigidly fastened together at Q and attached to two rigid supports at P and R. Shaft PQ is 8 cm in diameter and 2 m in length. Shaft QR is 6 cm in diameter and 1 m in length. A torque of $2 \times 10^5 \text{ N-mm}$ is applied at junction Q. Compute the maximum shearing stresses in each material. What is the angle of twist at the junction? Take modulus of rigidity for Aluminium as $0.3 \times 10^5 \text{ N/mm}^2$ and for Steel $0.9 \times 10^5 \text{ N/mm}^2$.



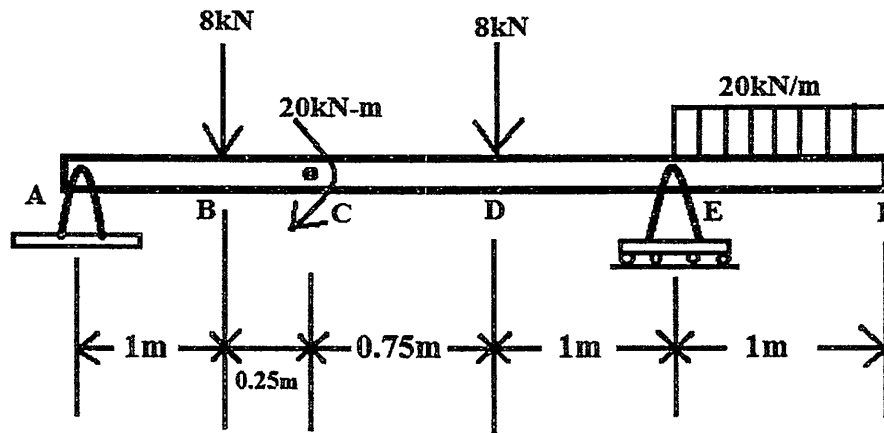
OR



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- V. Draw the shear force and bending moment diagrams for the beam shown below.

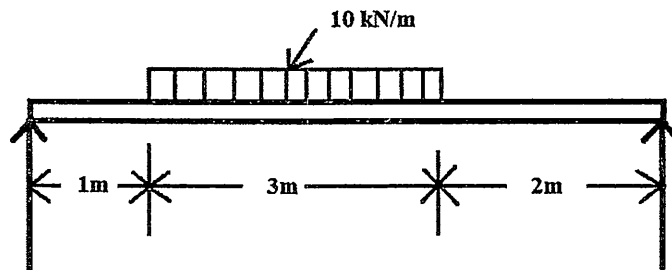


- VI. A cast iron water pipe 450 mm inner diameter and 25 mm thickness is supported at two points 10 m apart. Find the maximum stress in the metal when it is running full. Unit weight of C.I is 72 kN/m^3 .

OR

- VII. Discuss on various theories of failure and their applications to ductile and brittle materials.

- VIII. A beam of length 6 m is simply supported at its ends. It carries a uniformly distributed load of 10 kN/m as shown below. Determine the deflection of the beam at its mid-point and also the maximum deflection. Take $EI = 4.5 \times 10^8 \text{ N/mm}^2$.



OR

- IX. A mild steel tube is 75 cm in diameter and 0.25 cm thick. A short length of this tube is tested in compression and is found to yield at 500 N/mm^2 . The modulus of elasticity of the material of the tube is $2 \times 10^5 \text{ N/mm}^2$. A length of 2 m when tested as a strut with free-fixed end failed with a load of 180 kN force. Assuming the failing stress in Rankine formula to be the yield stress of the material, find the value of Rankine's constant 'a'. Find also the crushing load as by the Euler's formula.
